

Polychlorinated Biphenyl-77 Induces Adipocyte Differentiation and Proinflammatory Adipokines and Promotes Obesity and Atherosclerosis



Background Obesity, an inflammatory condition linked to cardiovascular disease, is associated with expansion of adipose tissue. Highly prevalent coplanar polychlorinated biphenyls (PCBs) such as 3,3',4,4'-tetrachlorobiphenyl (PCB-77) accumulate in adipose tissue because of their lipophilicity and increase with obesity. However, the effects of PCBs on adipocytes, obesity, and obesity-associated cardiovascular disease are unknown. Objectives In this study we examined in vitro and in vivo effects of PCB-77 on adipocyte differentiation, proinflammatory adipokines, adipocyte morphology, body weight, serum lipids, and atherosclerosis. Methods PCB-77 or 2,2',4,4,5,5'-hexachlorobiphenyl (PCB-153) was incubated with 3T3-L1 adipocytes either during differentiation or in mature adipocytes. Concentration-dependent effects of PCB-77 were contrasted with those of 2,3,7,8-tetrachlorodibenzo-p-dioxin (TCDD). For in vivo studies, we treated C57BL/6 wild-type (WT) or aryl hydrocarbon receptor (AhR)^{-/-} mice with vehicle or PCB-77 (49 mg/kg, by intraperitoneal injection) and examined body weight gain. In separate studies, we injected ApoE^{-/-} mice with vehicle or PCB-77 over a 6-week period and examined body weight, adipocyte size, serum lipids, and atherosclerosis. Results Low concentrations of PCB-77 or TCDD increased adipocyte differentiation, glycerol-3-phosphate dehydrogenase activity, and expression of peroxisome proliferator-activated receptor γ , whereas higher concentrations inhibited adipocyte differentiation. Effects of PCB-77 were abolished by the AhR antagonist α -naphthoflavone. PCB-77 promoted the expression and release of various proinflammatory cytokines from 3T3-L1 adipocytes. Administration of PCB-77 increased body weight gain in WT but not AhR^{-/-} mice. ApoE^{-/-} mice injected with PCB-77 exhibited greater body weight, adipocyte hypertrophy, serum dyslipidemia, and augmented atherosclerosis. Conclusions Our findings suggest that PCB-77 may contribute to the development of obesity and obesity-associated atherosclerosis.